RETURN ON INVESTMENT (ROI)

AURORA (AI-Powered Design Optimization, Compliance Validation & CAD–BOM Synchronization Platform)

1. Overview

Modern naval shipbuilding involves the design, validation, and management of millions of engineering data points across 3D CAD models, Bills of Materials (BOMs), production drawings, classification society rules, and multiple engineering disciplines. Maintaining consistency among these engineering artefacts throughout the vessel lifecycle is a significant challenge.

Manual verification of CAD models, BOMs, drawings, and regulatory compliance consumes substantial engineering effort while increasing the risk of inconsistencies, late-stage design corrections, manufacturing errors, procurement mismatches, and project delays. As ship complexity increases, these issues become increasingly expensive to identify and rectify.

AURORA addresses these challenges by providing an AI-assisted engineering intelligence platform that automatically synchronizes CAD models with Bills of Materials, validates compliance against classification society rules, detects inconsistencies before production, and recommends optimized engineering alternatives.

The return on investment generated by AURORA extends well beyond direct cost savings. The platform enhances engineering productivity, improves design quality, minimizes rework, accelerates approval cycles, strengthens configuration management, and creates a reusable digital engineering knowledge base that continues delivering value throughout future shipbuilding programmes.

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2. Sources of Economic Benefit

A. Reduction in Engineering Review Effort

Engineering teams currently spend significant time manually reviewing CAD assemblies, validating BOMs, checking production drawings, and ensuring compliance with design standards before every major engineering release.

AURORA automates a substantial portion of these verification activities through AI-assisted analysis and rule-based validation.

Expected Benefits

- Faster engineering reviews
- Reduced engineering man-hours
- Higher productivity of senior design engineers
- Faster design approval cycles
- Reduced dependence on manual checking
- Increased engineering throughput

B. Elimination of CAD–BOM Synchronization Errors

Differences between CAD assemblies and production Bills of Materials often remain undetected until procurement, fabrication, or assembly stages, resulting in expensive downstream corrections.

AURORA continuously synchronizes CAD models, BOMs, drawings, and engineering metadata to ensure complete configuration consistency throughout the design lifecycle.

Expected Benefits

• Significant reduction in BOM discrepancies

• Elimination of duplicate or missing components

• Improved procurement accuracy

• Reduced manufacturing errors

• Improved configuration control

• Better inventory planning

C. Automated Compliance Validation

Compliance verification against Classification Society rules, naval standards, and customer requirements is typically performed manually and requires extensive engineering effort.

AURORA automatically validates designs against configurable engineering rule libraries and identifies violations before submission.

Expected Benefits

• Faster compliance verification

• Reduced regulatory review effort

• Improved first-pass approval rates

• Lower certification costs

• Reduced engineering iterations

• Higher confidence in submitted designs

D. Reduction in Engineering Rework

Late identification of design inconsistencies frequently results in modifications after procurement, fabrication, installation, or production activities have already commenced.

AURORA identifies discrepancies during the digital design stage, allowing corrective actions before physical execution begins.

Expected Benefits

• Reduced fabrication rework

• Reduced installation rework

• Lower material wastage

• Improved manufacturing readiness

• Lower change implementation costs

• Reduced production disruptions

E. Improved Schedule Adherence

Engineering delays caused by repeated design reviews, compliance corrections, and BOM mismatches frequently impact downstream manufacturing schedules.

AURORA enables engineering teams to resolve issues significantly earlier in the design cycle, reducing approval delays and schedule slippages.

Expected Benefits

- Faster engineering release cycles
- Improved milestone achievement
- Better production planning
- Reduced project delays
- Improved resource utilization
- Increased schedule predictability

F. Improved Design Optimization

Traditional design optimization often depends on engineering experience and multiple manual iterations.

AURORA evaluates multiple design alternatives simultaneously using AI-assisted optimization techniques while satisfying engineering constraints and compliance requirements.

Expected Benefits

- Better-performing engineering designs
- Lower material utilization
- Reduced engineering iterations
- Improved weight optimization
- Lower lifecycle costs
- Improved design quality

G. Knowledge Preservation and Standardization

A significant portion of engineering knowledge exists only with experienced designers and subject matter experts.

AURORA captures engineering rules, historical design decisions, optimization strategies, and compliance knowledge into a reusable digital knowledge repository.

Expected Benefits

- Reduced knowledge loss
- Faster onboarding of new engineers
- Standardized engineering practices
- Improved organizational learning
- Consistent engineering decisions
- Continuous improvement across future programmes

3. Illustrative ROI Analysis

Current Situation

As highlighted in the problem statement, complex shipbuilding programmes routinely encounter:

• Manual CAD–BOM reconciliation

• Engineering design inconsistencies

• Multiple compliance review cycles

• Extensive engineering validation effort

• Configuration management challenges

• Procurement mismatches

• Manufacturing rework

• Schedule delays

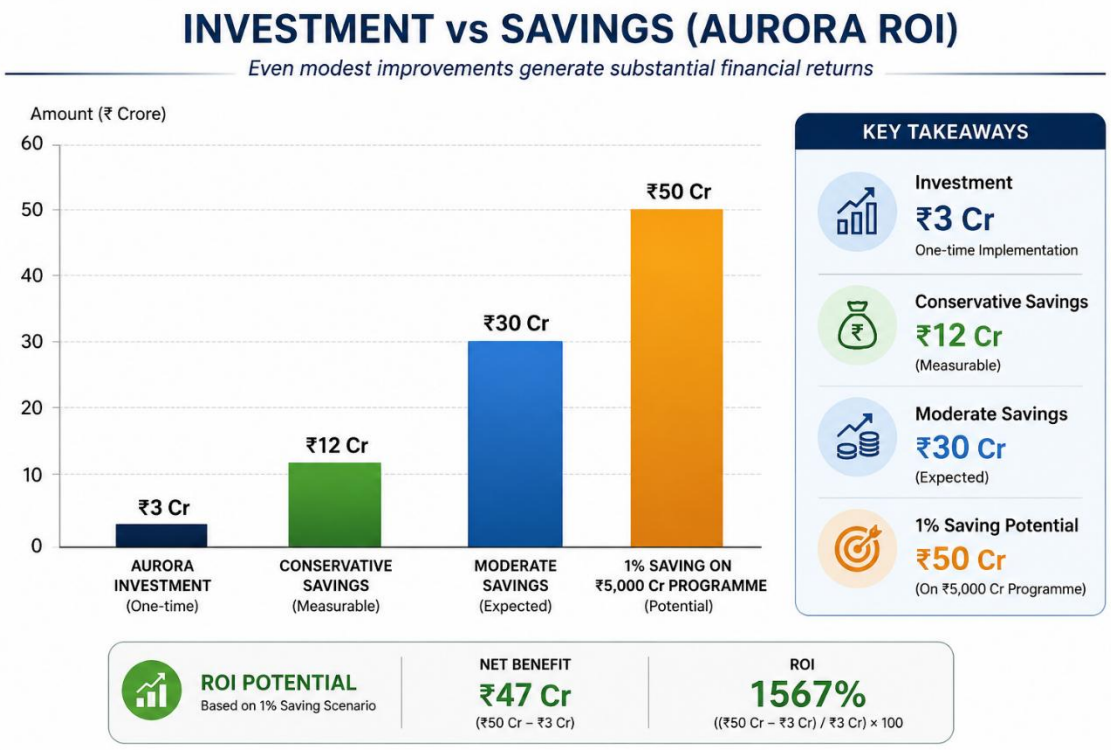
• Increased dependency on senior engineers

Even relatively small improvements in engineering accuracy and design efficiency can generate substantial financial savings throughout the lifecycle of a naval programme.

Consider a representative naval shipbuilding programme.

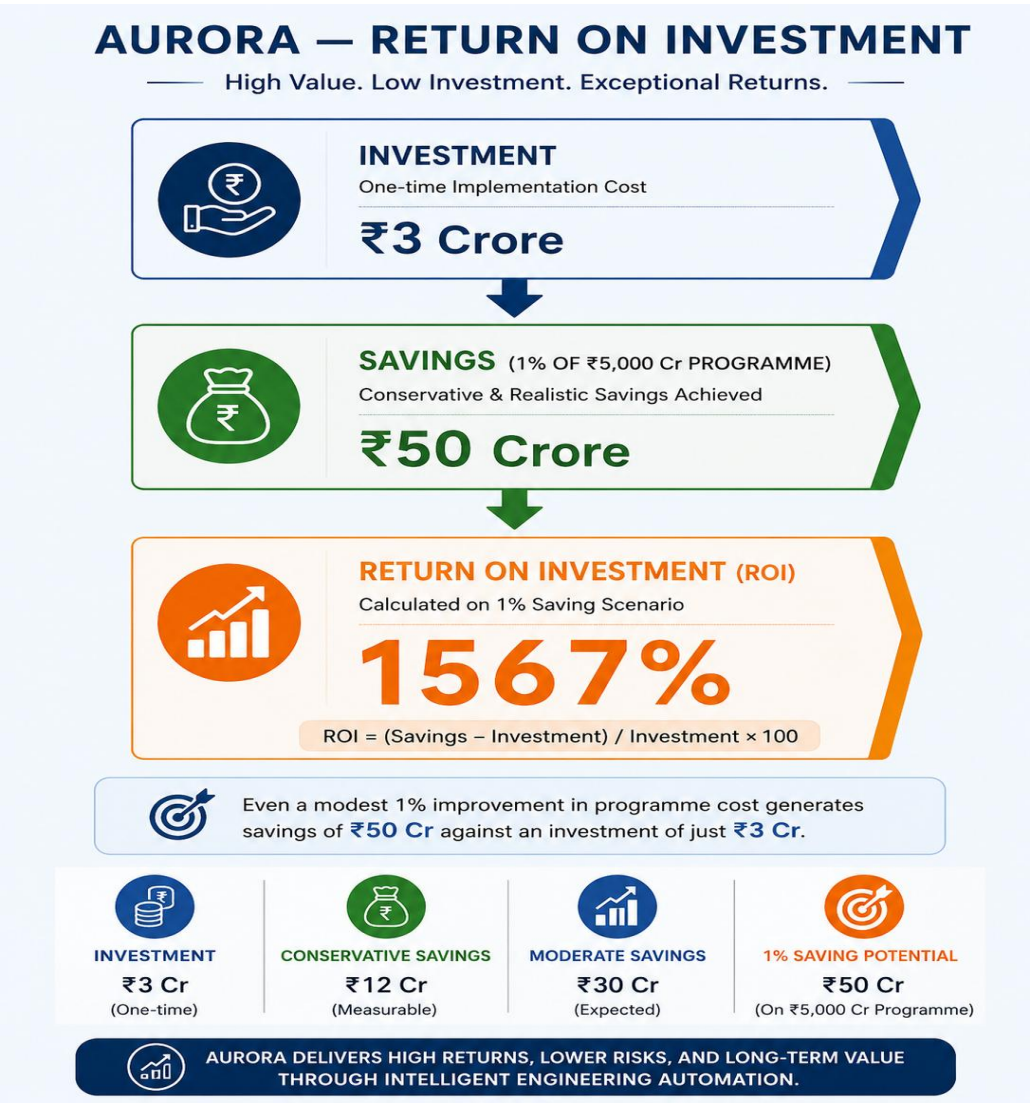
Parameter	**Value**
Typical programme value	₹5,000–10,000 Crore
Typical engineering-related rework	5–15% of engineering effort
AURORA implementation cost	₹3 Crore

Even modest improvements in engineering efficiency and design quality produce significant returns.



Scenario A – 1% Reduction in Programme Cost
For a ₹5,000 Crore programme:
Savings
= 1% × ₹5,000 Crore
= ₹50 Crore
Investment in AURORA
₹3 Crore
ROI Calculation
ROI = ((50 − 3) / 3) × 100
ROI ≈ **1567%**

Scenario B – Conservative Savings Case
Assume AURORA generates only **₹12 Crore** in measurable savings through:
• Reduced engineering review effort
• Lower CAD–BOM discrepancies
• Reduced compliance effort
• Reduced manufacturing rework
• Improved engineering productivity
Investment
₹3 Crore
ROI Calculation
ROI = ((12 − 3) / 3) × 100
ROI = **300%**



Additional Non-Financial Benefits

The measurable financial returns represent only part of the value generated by AURORA.

Additional strategic benefits include:

• Improved engineering quality

• Faster engineering approvals

• Better configuration management

• Improved auditability and traceability

• Higher design consistency

• Better collaboration between departments

• Enhanced digital transformation capability

• Long-term engineering knowledge preservation

ROI Summary

Item	Value
AURORA Investment	₹3 Crore
Conservative Savings	₹12 Crore
Moderate Savings	₹30 Crore
1% Saving on ₹5,000 Crore Programme	₹50 Crore

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4. Strategic Return Beyond Direct Cost Savings

While direct financial savings provide measurable ROI, the greater value of AURORA lies in transforming engineering processes into an intelligent, data-driven workflow.

Improved Engineering Quality

Automated validation significantly reduces human errors while improving consistency across engineering deliverables.

Reduced Programme Risk

Engineering discrepancies are identified before manufacturing begins, minimizing downstream project risks.

Increased Engineering Capacity

Engineers spend less time on repetitive verification activities and more time on innovation, optimization, and high-value engineering tasks.

Enhanced Digital Transformation

AURORA directly supports MDL's digital engineering roadmap by creating an integrated intelligence layer connecting CAD, BOM, drawings, compliance rules, engineering knowledge, and production planning.

Stronger Configuration Management

Continuous synchronization between engineering artefacts ensures accurate product configurations throughout the vessel lifecycle.

Better Regulatory Readiness

Automated compliance validation enables faster submissions to Classification Societies while reducing review iterations and certification delays.

Institutional Knowledge Retention

Engineering expertise becomes a reusable organizational asset rather than remaining dependent on individual engineers.

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5. Long-Term Value Creation

As additional projects are executed, AURORA continuously accumulates engineering knowledge, optimized design patterns, compliance rules, historical decisions, CAD metadata, and BOM intelligence.

Consequently, the value delivered by AURORA increases over time through:

• Improved optimization recommendations

• Expanded engineering rule libraries

• Better compliance coverage

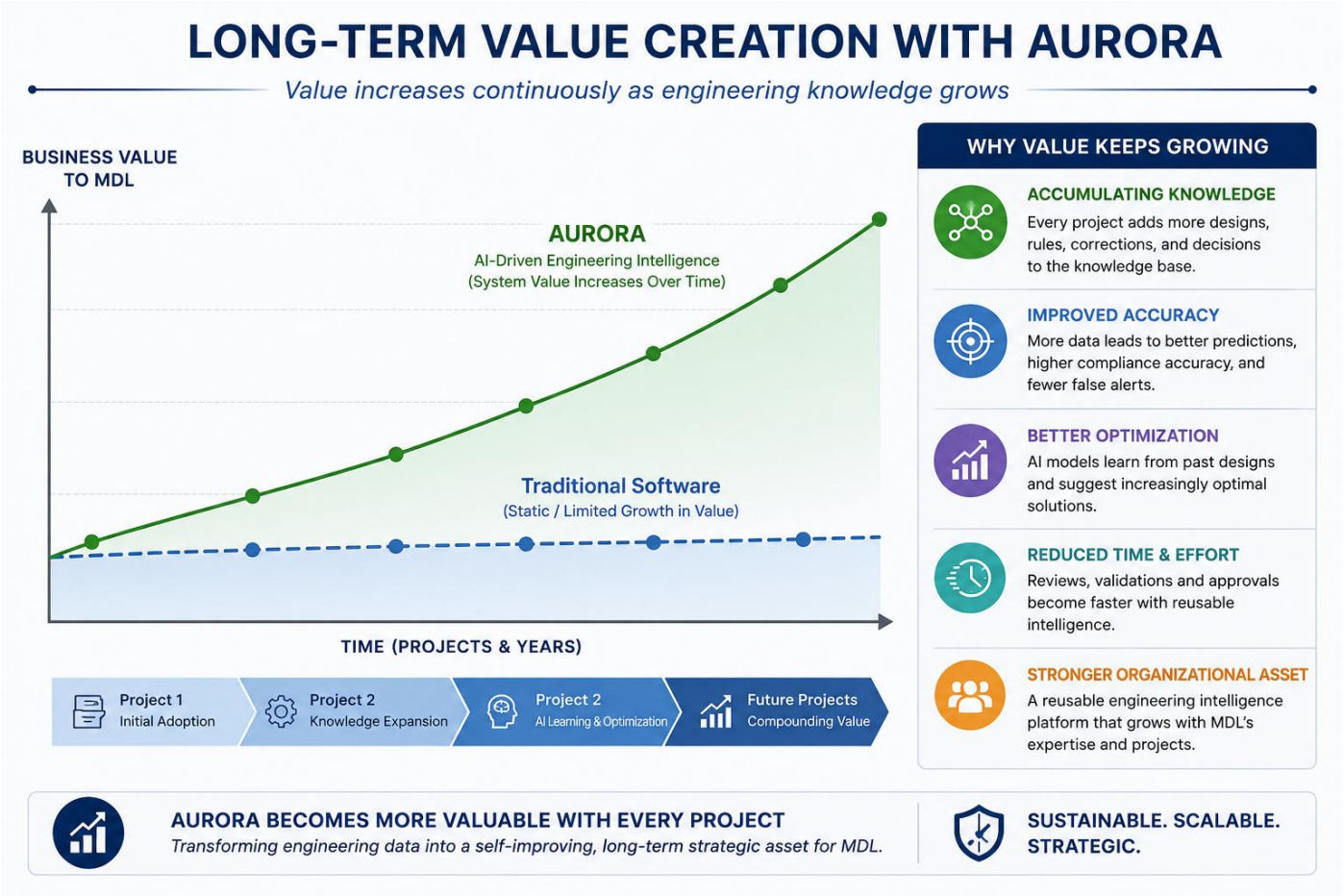
• Higher CAD–BOM synchronization accuracy

• Increased design reuse

• Enhanced engineering analytics

• Reduced dependence on individual experts

Unlike conventional engineering software whose value remains relatively static, AURORA becomes progressively more valuable as organizational engineering knowledge expands.



6. Conclusion

AURORA delivers return on investment through a combination of engineering productivity improvements, reduced design review effort, automated compliance validation, elimination of CAD–BOM inconsistencies, reduced manufacturing rework, improved schedule adherence, and long-term engineering knowledge preservation.

By enabling AI-assisted design optimization, automated engineering validation, and continuous synchronization of engineering artefacts, AURORA helps MDL deliver higher-quality ship designs faster, with greater confidence and significantly lower lifecycle costs.

For MDL, the true return is not measured solely in direct financial savings but in establishing a scalable engineering intelligence capability that strengthens digital shipbuilding, accelerates engineering decisions, improves programme execution, and supports future naval platforms with greater efficiency, consistency, and resilience.